ParkiScan Neuroverse · Game + Digital Twin · system architecture

Capture on the device, process locally, build one patient record, review it with a clinician, and feed the next session.

1 · CAPTURE
people and devices

Patient + avatar
Avatar and language chosen once; a persistent participant ID.

Webcam + microphone
Iris gaze, body pose, voice, claps. Processed on this computer; nothing is uploaded.

Wrist + ankle IMU
ESP32 + MPU-6050, 100 Hz, over USB serial to a local Python receiver that checks every reading.

avatar, language

camera, audio

UDP 127.0.0.1:47811

2 · UNITY 3D GAME
runs on this computer

Levels 1–3 + finale
Memory Lantern, Memory Core, spiral, nodes, mirror avatar, walking or seated drive.

On-device perception
Face Landmarker and BlazePose (ONNX, Unity Inference Engine, GPU).

Sensor input
Unity receives the checked IMU stream by UDP on 127.0.0.1:47811, uses it in the gait task and posts it to /sensors/imu.

Questionnaire + telemetry
Six check-ins, per-task metrics, quality flags and session files, all stamped with a session ID.

HTTP · JSON · localhost

3 · PARKISCANML SERVICE
Python · FastAPI · 127.0.0.1:8765

Sessions and features
Calls from the game: /session/start, /tasks, /features/voice, /sensors/imu, /session/stop. Extracts voice, spiral, cognition, gait, pose and gaze features and aligns the IMU windows.

Modality models
Daphnet freeze model (ankle IMU) and spiral model (experimental). The other models are research-only and are never applied to game data.

Multimodal assessment
Per-domain metrics with a data-quality state: valid, caveats, insufficient, invalid or not collected. The overall Parkinson's class is withheld until validated on clinician-labelled sessions.

session record, twin update

4 · RECORD, REVIEW, ACT
digital twin, dashboard, next steps

Patient record + digital twin
FHIR-style record (synthetic, SNOMED CT, LOINC), wearables on a 15-minute grid, levodopa PK/PD, Model A: freeze within 3 s, Model B: freeze within 2 h, what-if re-simulation. Game sessions arrive by /twin/ingest_game.

Clinician dashboard
In the game: F2, or the button after Level 3, live or as an offline snapshot. In a browser: /dashboard and a static copy for GitHub Pages.

Rehabilitation + next session
Proposed: clinician-set home activities, then re-assessed.

next session: assessment and rehabilitation loop (proposed)

Implemented and exercised in QA

Built; limited data or no live-hardware test

Proposed extension

All data stays on this computer (localhost only).

Model A predicts a freeze within 3 s from the ankle accelerometer (real Daphnet data, leave-one-subject-out). Model B predicts a freeze within 2 h from the record, wearables and the levodopa model (synthetic cohort).
Research prototype, not a medical device. Patient data are synthetic or anonymised open data.